

A photograph of four students in a library setting. A young man in a grey t-shirt is smiling and looking towards a young woman with glasses who is holding a laptop. Another young woman is looking at the laptop, and a fourth student is partially visible in the foreground. They are all gathered around a table with books and papers. The background shows bookshelves filled with books. The image has a semi-transparent blue overlay on the left side and a semi-transparent red overlay on the right side.

Iterating Collections

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- Modifying a collection while looping over it can lead to a *ConcurrentModificationException*.
- Iterators (interface *Iterator*) help but have no *add()* method.
- The *ListIterator* (for *Lists*) does have an *add()* method.
- The following table encapsulates (some) ways of inserting and removing for *Lists*, *Sets* and *Maps*.



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	Inserting	Deleting
	Note: Iterator does not have an add(E)	
List	ListIterator add(E) CopyOnWriteArrayList add(E) - concurrency	Iterator remove() Collection removeIf(Predicate)
Set	Set addAll(Collection) - deferred addition CopyOnWriteArraySet add(E) - concurrency	Iterator remove() Collection removeIf(Predicate)
Map	Set putAll(Collection) - deferred addition ConcurrentHashMap put(K, V) - concurrency	Iterator remove() Streams with a filter